Drafty

Drafty Executive Summary

Problem

Every winter, millions of renters open a utility bill that is far higher than expected and have no idea why. In older multifamily buildings, a large share of heating energy is lost through leaky windows, gaps under doors, unsealed outlets and poorly insulated walls. Renters usually pay the utility bill but do not own the building, so they are stuck. A professional energy audit typically costs several hundred dollars, is designed for homeowners, and often ends in a report that recommends upgrades a tenant cannot make. Small landlords, who own most rental units in the country, have the opposite problem: they are responsible for the building but rarely see the bills, so they have little data and little motivation to act.

We saw this ourselves. In a survey of 212 students and young renters at our university and nearby neighborhoods, 78 percent said their winter heating costs were higher than expected, and only 9 percent could name the cause. Residential energy costs also keep rising, which makes the problem sharper each year. The result is wasted money, wasted energy and avoidable emissions, with no simple tool to fix it.

Solution

Drafty is a sensor-and-app kit that maps heat loss in an apartment and tells the user what to fix first. Each kit has six small wireless sensors that measure temperature, humidity and air movement. The user places them at windows, doors, outlets and baseboards for about 48 hours and pairs them with the app. Our software combines the readings with outdoor weather and basic details about the unit, then produces a heat-loss map and a ranked list of fixes sorted by cost, effort and estimated payback.

Most recommended fixes cost under 40 dollars and need no contractor or permission: weatherstripping, door sweeps, outlet gaskets, window film, curtain changes. The app links to the parts and gives short installation guides. After fixes are made, users can rerun the sensors to see measured improvement, which is what makes Drafty different from a generic checklist. Landlords can also use a portfolio dashboard to compare units and prioritize repairs.

Market

The United States has about 44 million renter households. Most pay at least part of their own energy costs. We focus first on renters in older multifamily buildings in cold and mixed climates, roughly 12 million households. If a quarter of them bought one kit at our planned price, the one-time hardware market alone would be above 100 million dollars, before recurring landlord subscriptions. The second segment is small landlords, owners of one to twenty units, who hold a large share of rental housing and are underserved by existing energy tools. We estimate about 2 million such owners nationwide.

A third, longer-term segment is utilities and city energy programs, which are required in many states to meet efficiency targets and pay for measurable savings. Our anonymized data on where homes lose heat is valuable to them.

Business Model

We earn money in three ways. First, we sell the sensor kit directly to renters for 59 dollars. Our current cost of goods is about 24 dollars per kit at small volume, and we expect it to fall below 18 dollars when we order at scale. Second, we sell a landlord subscription at 6 dollars per unit per month, which includes the portfolio dashboard, repeat scans each season and a prioritized repair plan. Third, we plan to license anonymized, aggregated building data to utilities and efficiency programs. Affiliate revenue from recommended parts adds a small additional margin.

Sensors are reusable, so a single kit can be lent between units or reused each year, which lowers the cost for landlords and keeps subscription customers engaged.

Competition

Our competitors fall into three groups. Professional energy auditors give detailed results but cost hundreds of dollars and serve homeowners. Smart thermostats and general home sensors collect data but do not diagnose heat loss or recommend fixes. Free online checklists and utility tips are cheap but generic and do not tell a renter what is wrong in their particular apartment. Thermal-camera attachments for phones exist, but they need skill to interpret and only show surface temperature at one moment.

Drafty is built for renters and small landlords, costs a fraction of an audit, and gives a ranked, apartment-specific action list with measured before-and-after results. Our advantage is the combination of low-cost hardware with analysis software, and the data we collect with every scan will make our recommendations more accurate over time.

Team

Maya Okonkwo-Reyes, our CEO, is a senior in environmental policy who ran the renter survey and leads customer development and partnerships. Daniel Whitfield, our CTO, is a senior in electrical engineering who designed the sensor hardware and firmware. Priyanka Venkataraman, our head of data, is a master's student in data science who built the heat-loss model and the ranking engine. Tomas Brandt, our head of product, is a junior in industrial design who leads the app and kit design. We are advised by a faculty member in building energy systems and a retired property manager with thirty years of experience in small multifamily buildings.

Milestones and Financials

We have defined the core concept, built a working prototype of the kit and app, and put it in front of outside users. Over the last winter we ran Drafty in 41 apartments. Of these, 27 users paid for the kit at a discounted pilot price, and one local property manager with 60 units signed a paid pilot for 12 units. In follow-up scans, users who completed at least three recommended fixes measured an average 14 percent reduction in temperature loss at problem points, and self-reported bills dropped by an estimated 8 to 12 percent. Our app is live and our website collects signups, with a waitlist of over 400 people.

We have not yet formed a legal entity and plan to incorporate in the coming months. To date we have spent about 9,000 dollars, mostly on parts and prototypes, funded by pilot revenue and a small university grant. Pilot revenue so far is about 2,300 dollars. We project 85,000 dollars in revenue in the first full year, mostly from kit sales and landlord pilots, rising to roughly 420,000 dollars in year two as subscriptions grow and we add our first utility data pilot.

What We Are Asking For

We are seeking 40,000 dollars in prize or seed funding. About half would go to our first production run of 800 kits and certification testing, 30 percent to app development and our data pipeline, and the rest to legal formation, insurance and marketing to landlords. Beyond funding, we would value introductions to property management associations, utility program contacts and mentors with hardware manufacturing experience. With this support we expect to complete incorporation, double our landlord pilots, and be ready for a public launch ahead of next winter.